

A Geometric Diagnosis of Cross-Corpus Failure in Speech Emotion Recognition

Thesis Summary
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Abstract—Fine-tuned self-supervised speech models achieve high in-domain accuracy for speech emotion recognition (SER), yet their performance collapses when evaluated on a different corpus with an unchanged emotion taxonomy. This work diagnoses the geometric origin of such cross-corpus failure for a wav2vec 2.0 model fine-tuned on RAVDESS and evaluated on ESD. Through k -NN neighborhood analysis, linear probing, principal-angle analysis, subspace ablation, and Procrustes alignment validated on held-out speakers, the failure is localized in the classification head rather than in the encoder: emotional information survives in the frozen representation but along rotated axes. A compact head retrained on the frozen backbone absorbs this rotation, recovering accuracy from 19.73% to 75.12% on ESD and transferring zero-shot to German, Greek, and French corpora with gains of up to +78 macro-F1 points.

I. CONTEXT AND OBJECTIVES

Cross-corpus failure is the phenomenon whereby a SER model reaching high accuracy on its fine-tuning dataset suffers a severe performance degradation when evaluated on a different corpus, even under an identical emotion taxonomy. Rather than treating the model as a black box and retraining it end to end, this thesis decomposes the model into its encoder and its classification head and asks *where* the failure originates, studying the geometry of the latent space via linear probing and subspace analysis, and deriving corrective strategies consistent with the diagnosis.

A. Experimental setup

The baseline is ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition: a wav2vec 2.0 XLSR-53 encoder, pretrained self-supervised on $\approx 56,000$ hours across 53 languages without emotional annotation, fine-tuned on RAVDESS (1,440 clips, 24 English speakers, 8 emotions) with 82.23% validation accuracy. The target corpus is ESD (35,000 utterances, 20 speakers, English and Mandarin, 5 emotions). All analyses operate on frozen embeddings obtained by statistics pooling of the last Transformer layer, $\mathbf{z} = [\boldsymbol{\mu} \parallel \boldsymbol{\sigma}] \in \mathbb{R}^{2048}$.

B. The failure and two hypotheses

Evaluated zero-shot on ESD, the baseline collapses to 19.73% accuracy (18.38% macro-F1), near the 20% chance level: predictions concentrate on *Angry/Happy/Surprised, Sad* is almost never predicted (74/7,000), and high-arousal classes are mutually confused. Two hypotheses frame the diagnosis. **H1 (representation)**: the encoder genuinely loses emotional information on out-of-distribution data, so no classifier could succeed. **H2 (alignment)**: the information is preserved, but the decision boundaries learned on the source domain are misaligned with the target embedding distribution.

II. DIAGNOSTICS

A. Local structure of the latent space

A k -NN probe ($k = 15$) on frozen ESD embeddings tests whether local neighborhoods already predict emotion: Intra-speaker classification accuracy reaches $80.29\% \pm 7.21$, and decreases to $70.04\% \pm 5.46$ when the latent representations of all speakers are jointly considered. The drop suggests that language, gender, and speaker identity dominate local proximity, motivating a global analysis.

B. Linear probing (H1)

A speaker-independent multinomial logistic probe on frozen ESD embeddings reaches 77.64% accuracy (roughly 58 points above the baseline head on the same data) showing that emotion remains linearly separable. Attribute probes reveal a clear hierarchy on ESD: language (99.84%), speaker (96.01%), and gender (88.13%) are all more separable than emotion (77.64%). Per-class behaviour is non-uniform: *Angry, Neutral*, and *Sad* separate well (F1 83–88%), while *Happy* (62.4%) and *Surprised* (69.8%) are confused mainly with each other.

C. Subspace independence and ablation

Principal angles between the emotion subspace $\mathcal{E} = \text{span}(W_{\text{emo}})$ and the auxiliary subspaces, $\cos(\theta_i) = \sigma_i(U_{\mathcal{E}}^T U_{\mathcal{A}})$, all exceed 75° on ESD (language 85.03° , gender 75.99° , speaker 75.10°): affective directions are near-orthogonal to the confounds. Projecting embeddings onto the

orthogonal complement of each attribute subspace, $\tilde{\mathbf{z}} = (I_d - U_{\text{att}}U_{\text{att}}^\top)\mathbf{z}$, and retraining the emotion probe leaves accuracy essentially unchanged (77.20% with all confounds removed vs. 77.64%): the residual overlap carries no emotional signal. **The strong form of H1 is rejected.**

D. Procrustes alignment (H2)

If emotion is preserved but reoriented, a single orthogonal transformation should align the source and target decision boundaries. The optimal rotation has closed form

$$R^* = \arg \min_{R^\top R = I_d} \|\hat{W}^{\text{RAV}} R - \hat{W}^{\text{ESD}}\|_F = UV^\top,$$

with $U\Sigma V^\top = \text{SVD}((\hat{W}^{\text{RAV}})^\top \hat{W}^{\text{ESD}})$. Validation is speaker-independent: R^* is estimated on 16 ESD speakers, and the 4 held-out speakers are rotated into the RAVDESS frame and classified by the *frozen* RAVDESS probe (Table I).

TABLE I
SPEAKER-HELD-OUT VALIDATION ON ESD (5 RANDOM SPLITS).

Condition	Accuracy (%)	Macro F1 (%)
Baseline (no rotation)	41.9 $\pm$ 2.1	37.8 $\pm$ 1.5
Procrustes (rotated)	71.0 $\pm$ 2.1	70.5 $\pm$ 2.8
Oracle (ESD-native probe)	75.2 $\pm$ 3.5	75.2 $\pm$ 3.6

The recovery (41.9% $\rightarrow$ 71.0%, against a 75.2% oracle) occurs on speakers disjoint from the estimation of R^* : the rotation transfers the emotional geometry to an unseen group. **H2 is confirmed:** the dominant failure mode is a misalignment of the classification head, implying that retraining only the head should suffice.

III. RETRAINING AND ZERO-SHOT GENERALIZATION

A compact MLP head (LayerNorm, $1024 \rightarrow 256 \rightarrow 128 \rightarrow 5$, GELU, dropout 0.1) is trained on the frozen encoder with ESD labels under five-fold speaker-independent cross-validation, stratified by language. It recovers 75.12% $\pm$ 3.4 accuracy (75.00 $\pm$ 3.7 F1), within three points of the 77.64% linear-probe bound: gradient descent implicitly absorbs the closed-form Procrustes rotation.

The retrained head (best fold, encoder frozen) is then tested in pure zero-shot on three corpora never seen in any supervised stage, restricted to shared emotion classes (Table II).

TABLE II
ZERO-SHOT TRANSFER TO UNSEEN LANGUAGES (%).

Corpus	Setup	Accuracy	F1	Δ F1
EmoDB (DE)	Baseline	21.24	16.48	–
	New head	94.69	94.63	+78.15
AESDD (EL)	Baseline	61.16	49.49	–
	New head	74.93	75.00	+25.51
CaFE (FR)	Baseline	14.04	13.23	–
	New head	63.43	60.84	+47.61

Improvements are obtained in every out-of-distribution domain, with diagonally dominant confusion matrices in all cases.

IV. CONCLUSIONS AND FUTURE WORK

The cross-corpus failure of a fine-tuned wav2vec 2.0 SER model was localized through a single line of reasoning from diagnosis to intervention: emotional information remains linearly accessible on the target corpus, is encoded near-orthogonally to language, gender, and speaker identity, and differs from the source configuration essentially by a rigid rotation of the decision boundaries. Retraining a compact head on the frozen encoder suffices to recover in-domain performance and enables substantial zero-shot cross-lingual transfer.

Future work moves beyond discrete emotion labels toward a continuous analysis of the latent space, mapping interpretable acoustic descriptors (eGeMAPS, e.g. spectral flux) directly onto the latent topology: low-intensity regions align with *Sad/Neutral* clusters and high-activation regions with *Angry/Happy/Surprised*, enabling an extended interpretability that isolates the physical causes of the latent organization of emotions.